MILITARY SPECIFICATION SHEET

TRANSFORMER, POWER, STEP-DOWN TF5V03ZZ

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the transformer described herein shall consist of this document and the latest issue of Specification MIL-T-27.

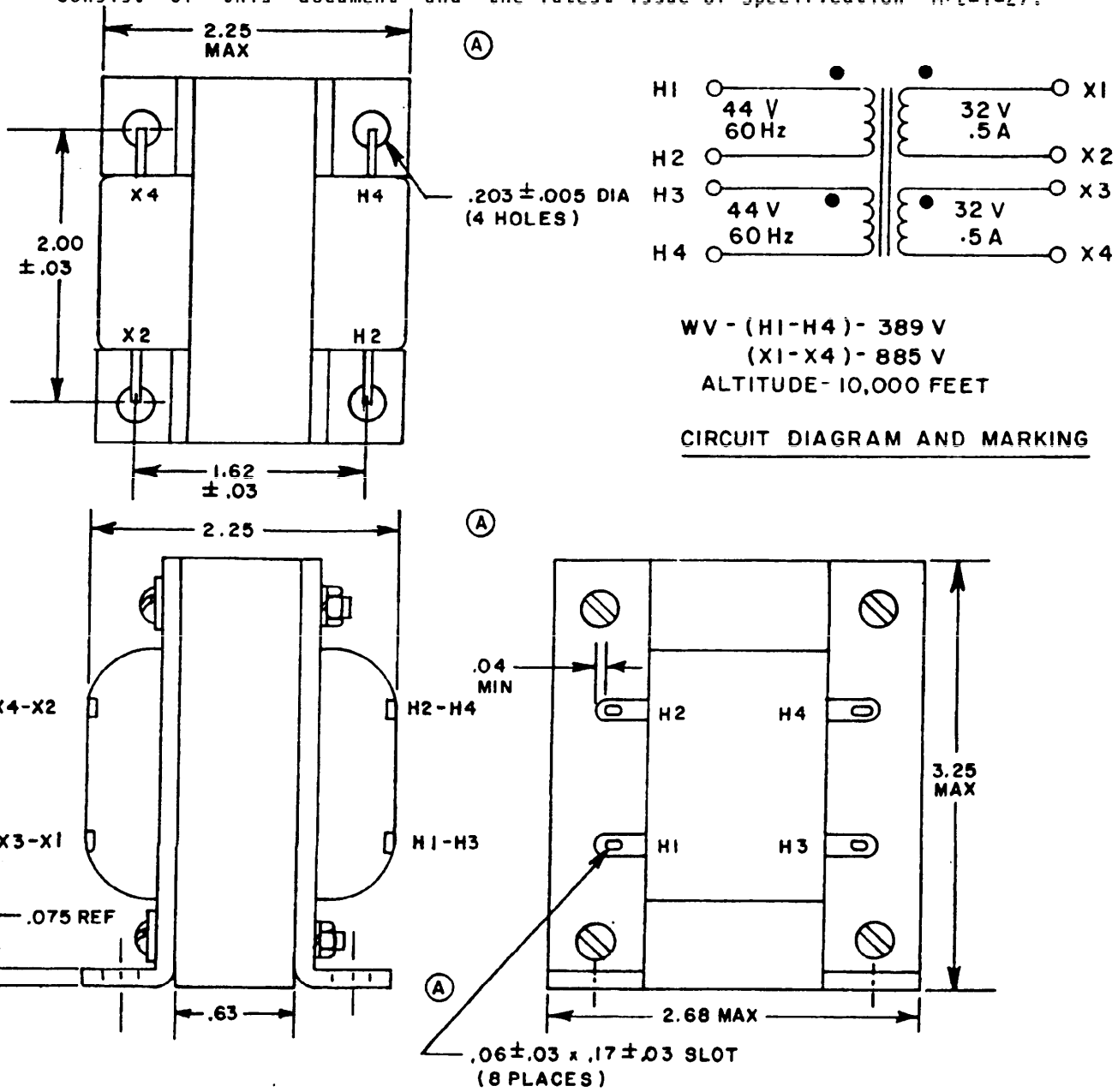


FIGURE 1. Dimensions and configuration.

Ⓐ denotes changes

FSC 5950

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INCHES	MM
.005	.13
.03	.8
.04	1.0
.06	1.5
.075	1.90
.17	4.3
.203	5.16
.63	16.0
1.62	41.1
2.00	50.8
2.25	57.2
2.68	68.1
3.25	82.6

Ⓐ

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Marking shall be on the side of the case.
4. Unless otherwise specified, tolerances are ±.005 (.13 mm) for three place decimals and ±.03 (.8 mm) for two place decimals.

FIGURE 1. Dimensions and configuration - Continued.

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REQUIREMENTS: (When numbers in parentheses, i.e., (H1-H2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings:

Primary voltage: (H1-H4): 88 Vrms, 60 hertz.

Primary current: (H1-H4): .5 ampere rms.

(A) Volt ampere: 22 volt ampere.

Secondary:

(X1-X2): 32 volts, .5 ampere rms.

(X3-X4): 32 volts, .5 ampere rms.

Working voltage:

(H1-H4): 389 volts, maximum.

(X1-X4): 885 volts, maximum.

Design and construction:

Dimensions and configuration: See figure 1.

Duty cycle: Continuous.

Case: Encapsulated.

Material: Epoxy.

Terminals: Solder type.

Material: Cadmium plated brass or tinned.

Height: .25 inch, minimum.

Weight: 28.6 ounces, maximum.

Operating temperature range: +5° to +155°C.

Terminal strength: MIL-STD-202, method 211, test condition A, 2.0 pounds.

Dielectric withstanding voltage (at sea level): 2,240 Vrms, except H1 to H3 1,090 Vrms.

Electrical characteristics:

No load: With 44 volts, 60 hertz across (H1-H2) or (H3-H4), current shall not exceed 100 milliamperes, and the power shall not exceed 8.0 watts.

Voltage across (X1-X2) and (X3-X4): 43 volts ±1.0 percent.

Rated load: With 44 volts, 60 hertz across (H1-H2) and (H3-H4) and a .5 ampere resistive load in (X1-X2) and (X3-X4) the voltage across (X1-X2) and (X3-X4) shall be 32 volts ±5 percent.

DC resistance:

(H1-H2), (H3-H4): 8.7 ohms ±20 percent.

(X1-X2), (X3-X4): 13.6 ohms ±20 percent.

Polarity: Additive, with terminals H2 and H3 connected, or terminals H2 and X1 or terminals H2 and X3.

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Temperature rise: 90°C maximum with 88 volts, rms, 60 hertz across (H1-H4) with H2 and H3 connected.

Shock (high impact): MIL-STD-202, method 207.

Marking location: See figure 1.

Part number: M27/296-01.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - SH, OS
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AR, WC
Navy - MC, AS
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-T-27/296A TRANSFORMER, POWER, STEP-DOWN

TF5VQ3ZZ

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